GreenRoute AI: An Explainable, Human-in-the-Loop System for Green Solvent Substitution and Synthesis Optimisation

Team InNOChem

4 participants

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Abstract

This document presents **GreenRoute AI**, a prototype intelligent system submitted to the InNOChem Hackathon. The solution addresses two thematic axes: (2.1) substitution of toxic solvents and (2.2) optimisation of synthesis routes. It satisfies all mandatory requirements: at least two capabilities (preventive, optimisation), full scientific transparency (XAI), strict Human-in-the-Loop, controlled LLM usage, and real engineering (Python, RDKit, ML, Docker). The system assists chemists in ranking solvent alternatives and reaction pathways using green metrics (E-factor, atom economy, biodegradability, VOC) while requiring explicit human validation for every critical decision. This document details the architecture, how each rule is met, deliverables, and task division among the four team members.

1 Problem Statement and Chosen Axes

The hackathon's central question is: *"How to design intelligent systems that help develop safer, greener, more efficient chemical processes with transparency and human control?"*

We focus on two complementary axes:

- **2.1 – Substitution of toxic and polluting solvents**: Replace harmful solvents (e.g., DMF, dichloromethane) with green alternatives without compromising performance.
- **2.3 – Optimisation of synthesis routes**: Minimise waste (E-factor), maximise atom economy, reduce energy consumption.

By combining both, GreenRoute AI proposes holistic improvements: changing a solvent often unlocks a greener synthesis path, and vice versa.

Capability	How we satisfy it
Preventive	Before any experiment, the system flags solvent toxicity, bioaccumulation, VOC emissions, and predicted waste.
Optimisation	For a target molecule, it ranks synthetic routes + solvent pairs by green metrics, yield prediction, and energy demand.

Table 1: The two mandatory capabilities are fully met (analytical capability is provided as a bonus).

2 Capabilities Covered (Mandatory 2)

3 How We Satisfy Every Golden Rule

3.1 Scientific Transparency (XAI) – 30% of score

- Every recommendation includes a plain-language justification: *"Cyclopentyl methyl ether is preferred over toluene because its E-factor is 0.5 lower, it is readily biodegradable (OECD 301F), and it reduces VOC emissions by 60%."*
- Complete traceability: all data sources (PubChem, ECHA, or simulated) are displayed.
- Uncertainty quantification: the model outputs a confidence interval for predicted yields and flags missing data (e.g., "No ecotoxicity data → conservative default used").
- No black-box: ranking uses a transparent weighted sum of green metrics; weights are user-adjustable.

3.2 Human-in-the-Loop (HITL) – Absolute Golden Rule

The AI proposes, the human validates. Workflow:

1. Chemist inputs a target molecule (SMILES or name).
2. System returns top-3 synthesis routes + solvent options with explanations.
3. An explicit **validation button** appears – the chemist must click "Approve for experiment" before any recommendation is considered final.
4. The chemist can override parameters (e.g., force a specific solvent, set max temperature).

This mechanism is hard-coded in the prototype and cannot be bypassed.

3.3 Controlled Use of LLMs – No Hallucinations

- LLM is used **only for auxiliary tasks**: natural language query parsing and explanation templating.

- The core decision engine (solvent ranking, yield prediction, green scoring) is a Random Forest model + rule-based system – **no LLM involvement** in scientific decisions.
- Every LLM output is verified against a solvent/reaction ontology; mismatches trigger fallback answers.
- **Result:** Zero uncontrolled hallucinations → no 10 penalty.

3.4 Real Engineering – No No-Code, No Simple Wrapper

- **No no-code tools:** All logic in Python (Flask, Pandas, RDKit, scikit-learn).
- **No simple LLM wrapper:** LLM is non-critical; core is trained ML + deterministic calculations.
- **Real engineering:** Data pipeline (Airflow optional), vector database (FAISS) for solvent fingerprints, REST API, uncertainty quantification (Monte Carlo dropout).
- **No pseudo-science:** Every green metric is clearly defined (see separate concepts document). All calculations are reproducible via Jupyter notebooks.

4 Technical Architecture (Summary)

```
[Web UI] → [Flask API] → [Orchestrator]
                                   Solvent DB (SQLite + FAISS)
                                   ML Model (Random Forest + RDKit descriptors)
                                   Green Metrics Calculator
                                   Explanation Engine (template-based)
```

Dockerised for reproducibility. The HITL state machine resides in the orchestrator.

5 Deliverables Status

Livrable	How we will provide it
6.1 Prototype (MVP)	Functional web app (live demo). Chemist enters a molecule → gets ranked solvents/routes → validation button. Not a visual mockup.
6.2 Architecture	PDF diagram (components, data flow, interfaces) + description.
6.3 Scientific note	Detailed: datasets (source or simulation declaration), biases, model limits, justification of green metrics, XAI method, HITL workflow.
6.4 Pitch	10 min presentation + 5 min Q&A. Slides include problem, solution, live demo, architecture, rules compliance, impact, team division.

6 Task Division for 4 Team Members

Role	Tasks
Participant 1 – Chemistry & Data Lead	Curate/simulate solvent greenness data (toxicity, E-factor, biodegradability). Define green metrics formulas. Write scientific note. Validate chemical plausibility.
Participant 2 – AI/ML Engineer	Build yield prediction model (RDKit descriptors + Random Forest). Implement uncertainty quantification. Develop solvent similarity search. Create ranking algorithm.
Participant 3 – Backend & Data Pipeline	Develop Flask API. Integrate ML model, DB, HITL state machine. Set up Docker. Ensure reproducible pipelines. Produce architecture diagram.
Participant 4 – Frontend & UX / Pitch	Build web interface (Streamlit/React). Implement validation button and explanation display. Create pitch deck. Coordinate live demo.

Table 2: Each member also participates in the final pitch (their own part).

7 Why This Will Reach the Podium

- **For the software developer:** Clean, modular Python + Docker + vector DB + uncertainty quant. No no-code, no wrapper.
- **For the chemistry expert:** Real green metrics (E-factor, atom economy, VOC, biodegradability). Actionable recommendations. HITL respects lab workflows.
- **For the AI expert:** Explainable AI (XAI), uncertainty communication, no hallucinations. Real ML (Random Forest) on cheminformatics descriptors.
- **Red flags avoided:** No no-code penalty, no wrapper penalty, no weak engineering, no pseudo-science, data justified, no hallucinations.
- **Bonus:** Use of RDKit (computational chemistry library) → +5 points. Real public dataset (e.g., ECHA or USPTO) → additional +5.

8 Conclusion

GreenRoute AI directly answers the hackathon's problem statement while strictly adhering to the golden rules. It augments chemists, replaces no human decision, and promotes verifiable green chemistry. With the proposed division of work and technical rigour, our team is positioned for a top finish.